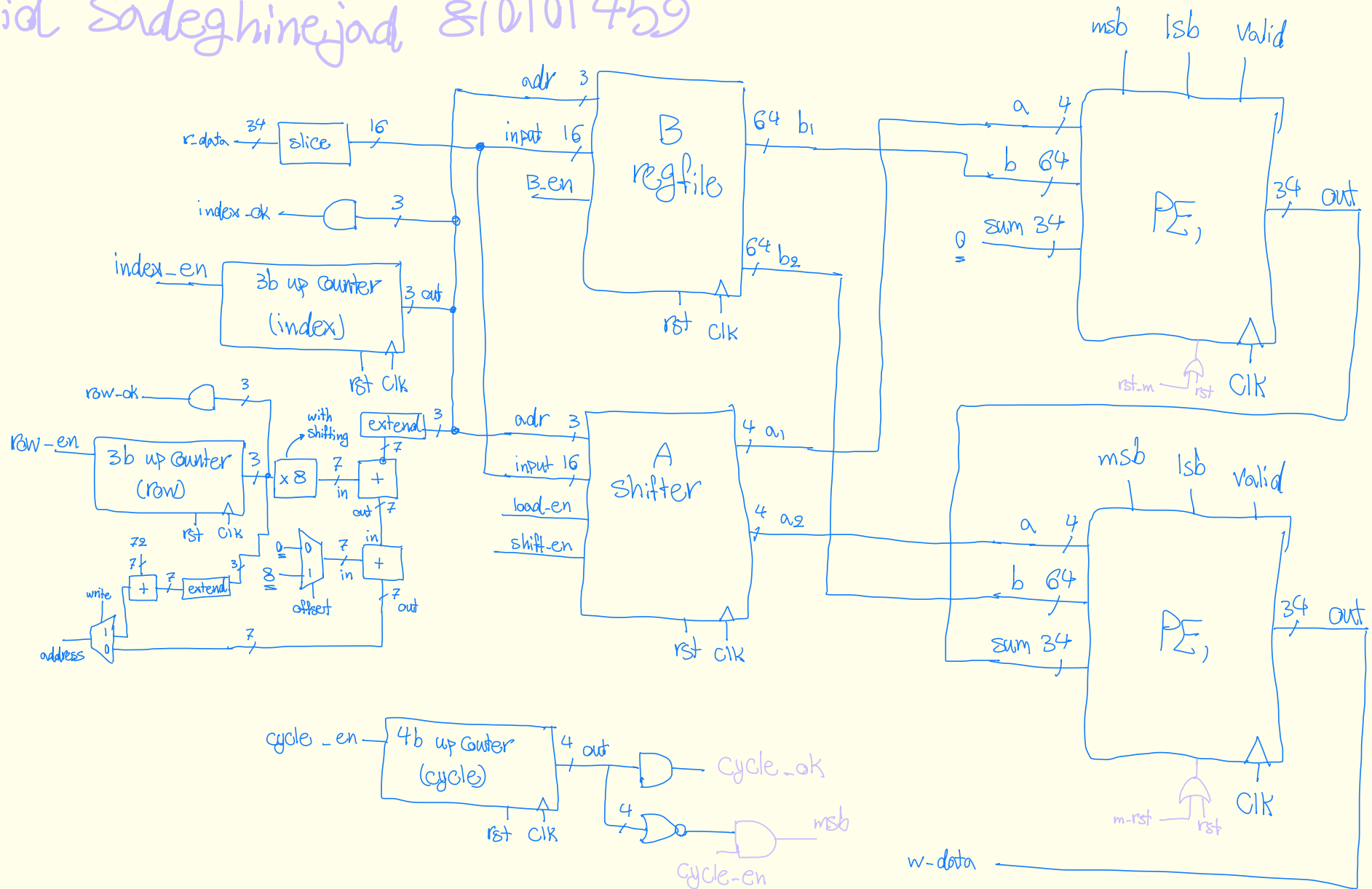


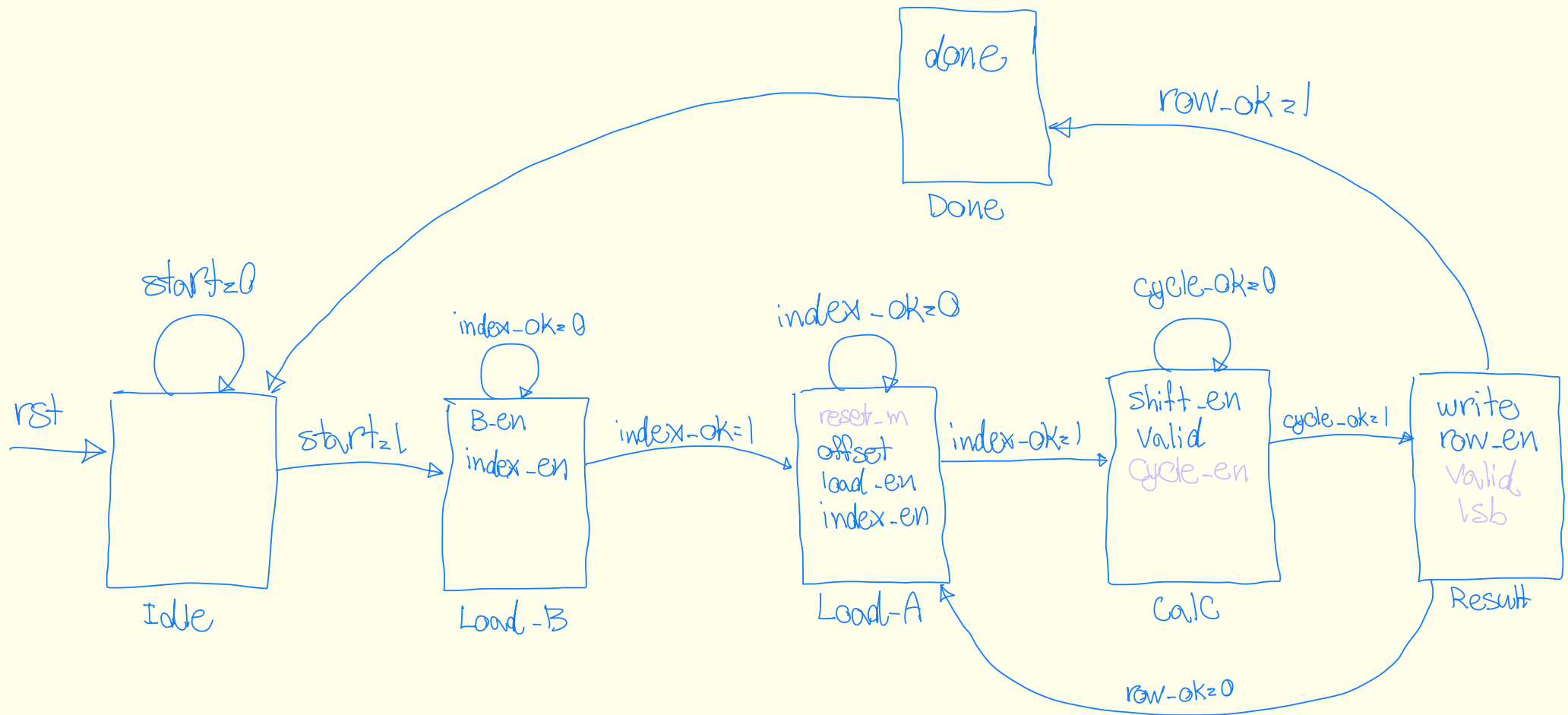
Matrix-vector multiplication Data path

Sepehr Bazarvan 810102303

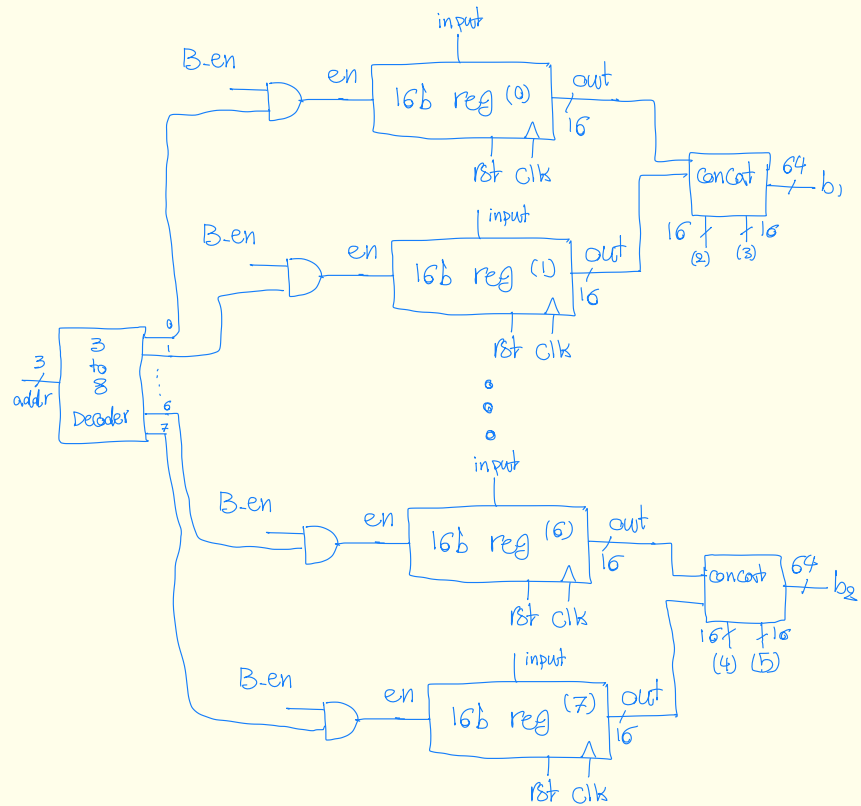
Majid Sadeghinejad 810101459



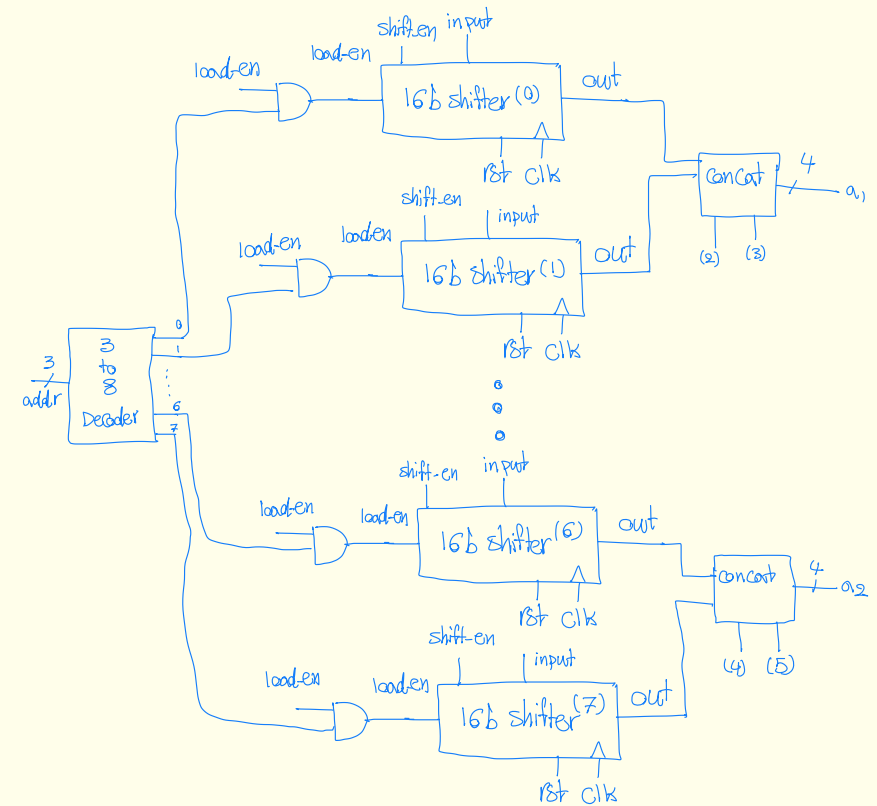
Matrix-vector multiplication Controller



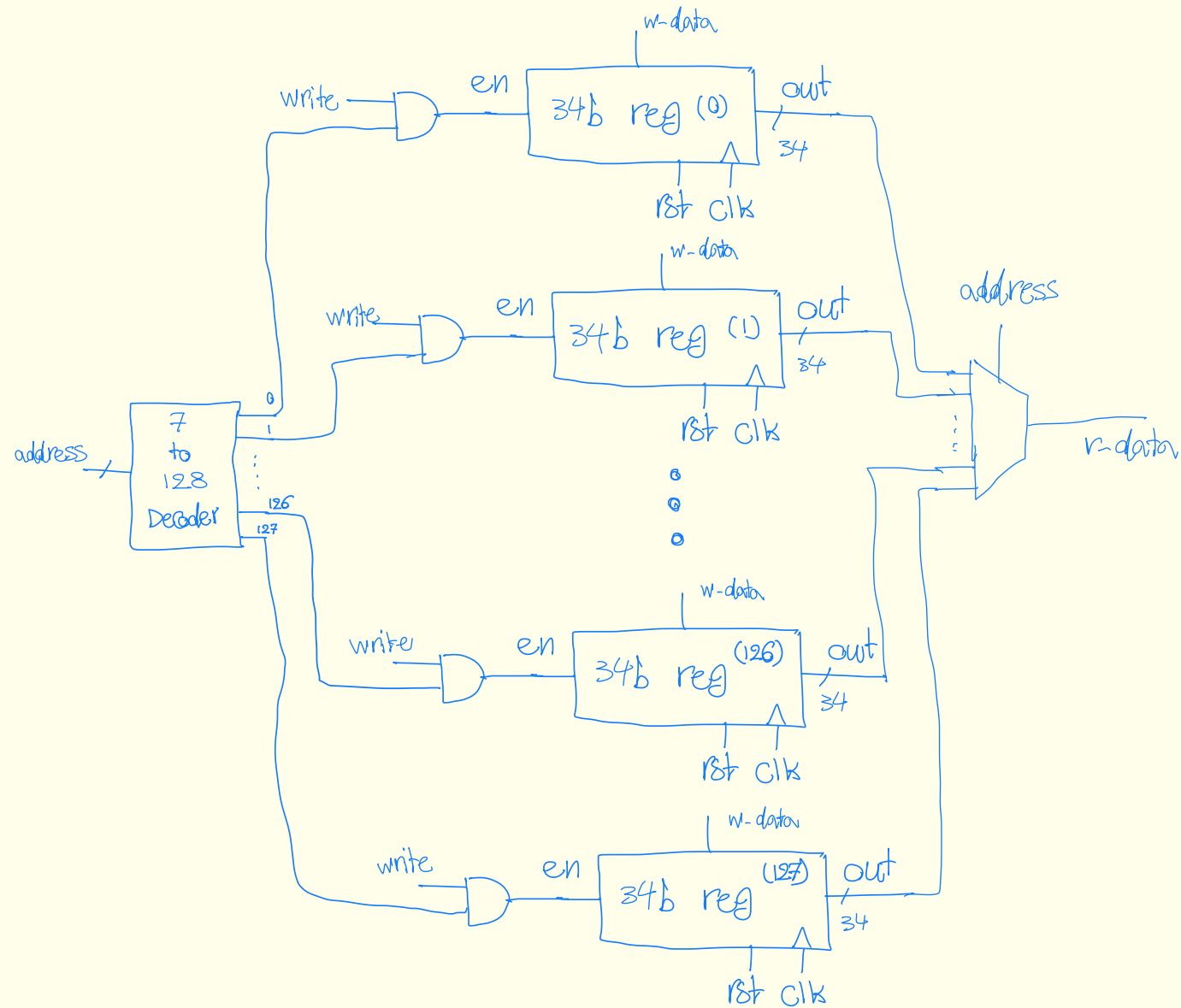
B-reg file



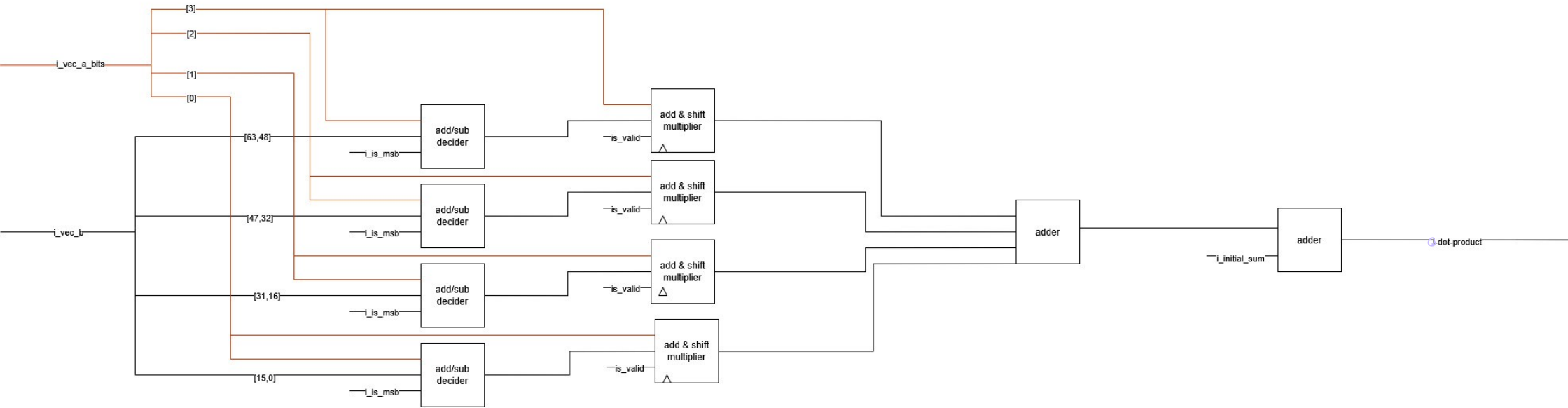
A-shifter



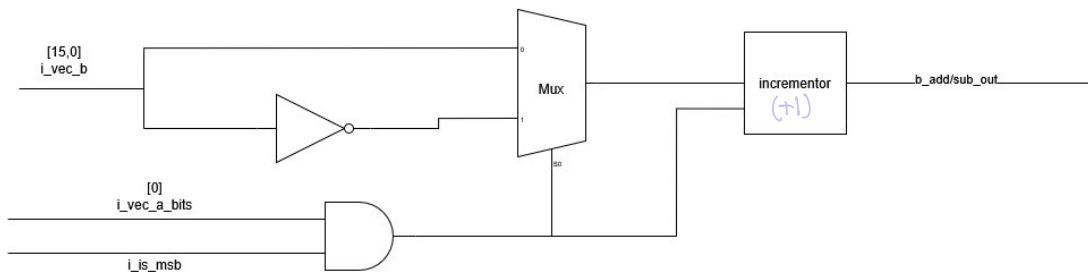
Memory



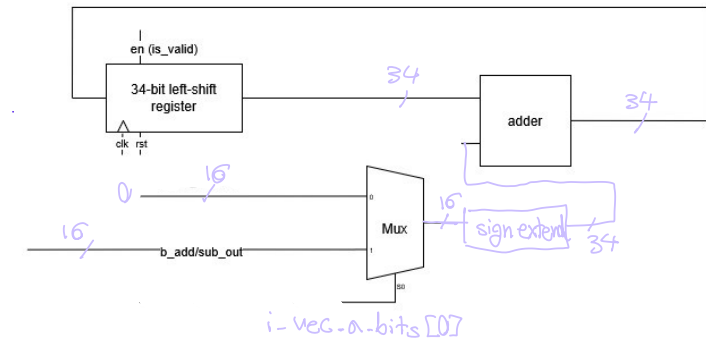
Stripes PE



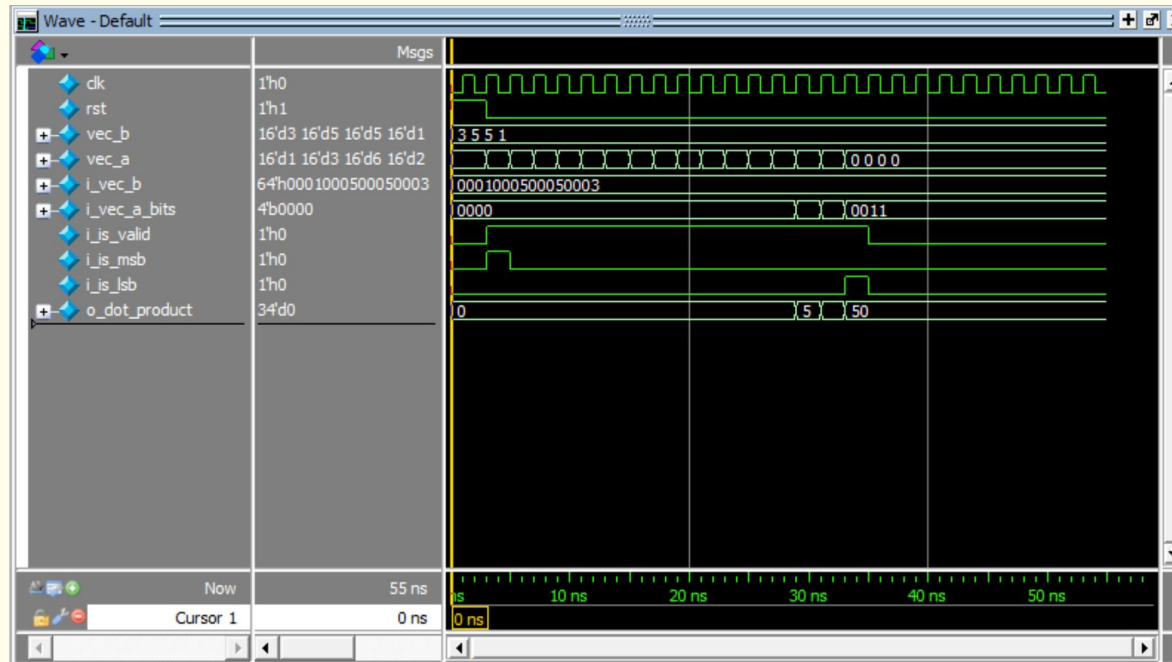
add/sub decoder



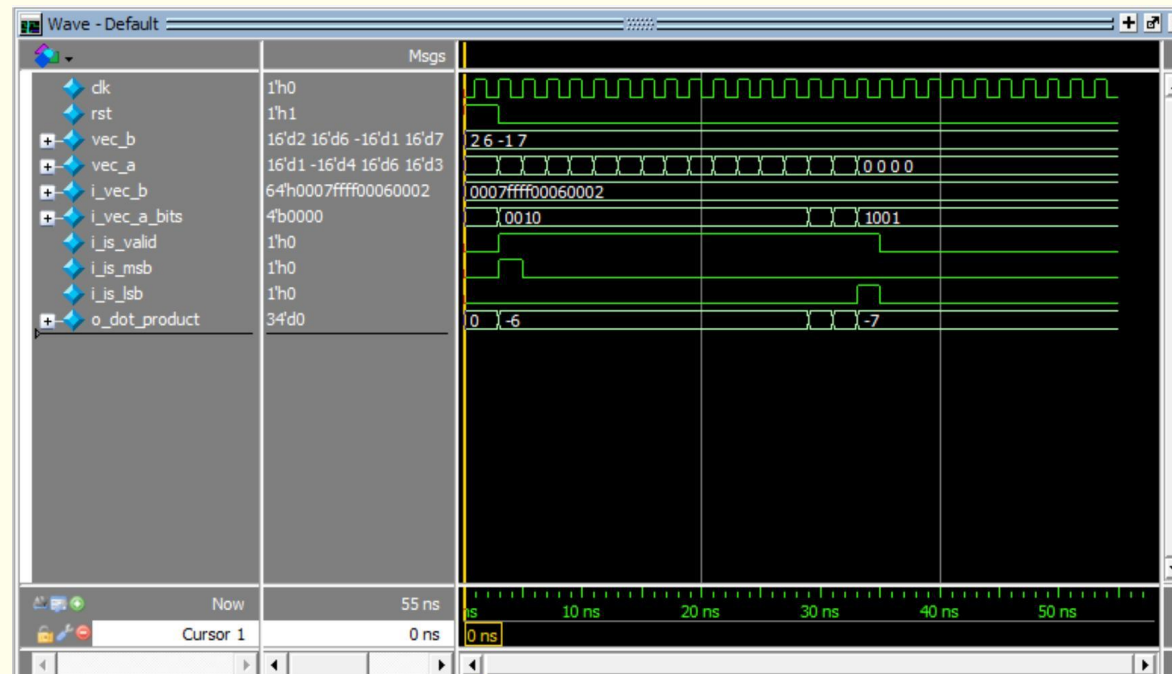
add & shift multiplier



Results (stripes-pe)



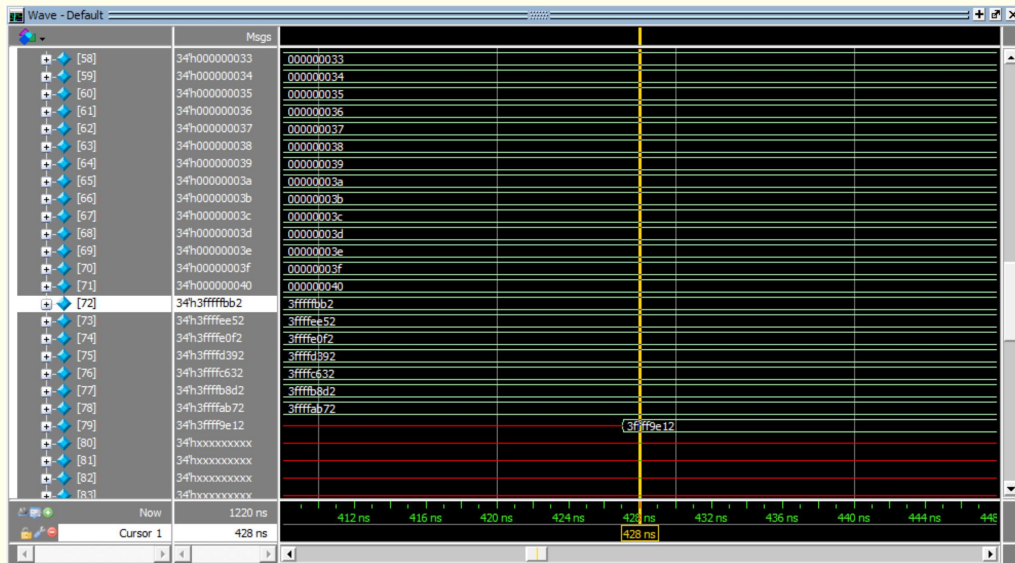
$$\begin{pmatrix} 1 \\ 3 \\ 6 \\ 2 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \\ 5 \\ 1 \end{pmatrix} = 50$$



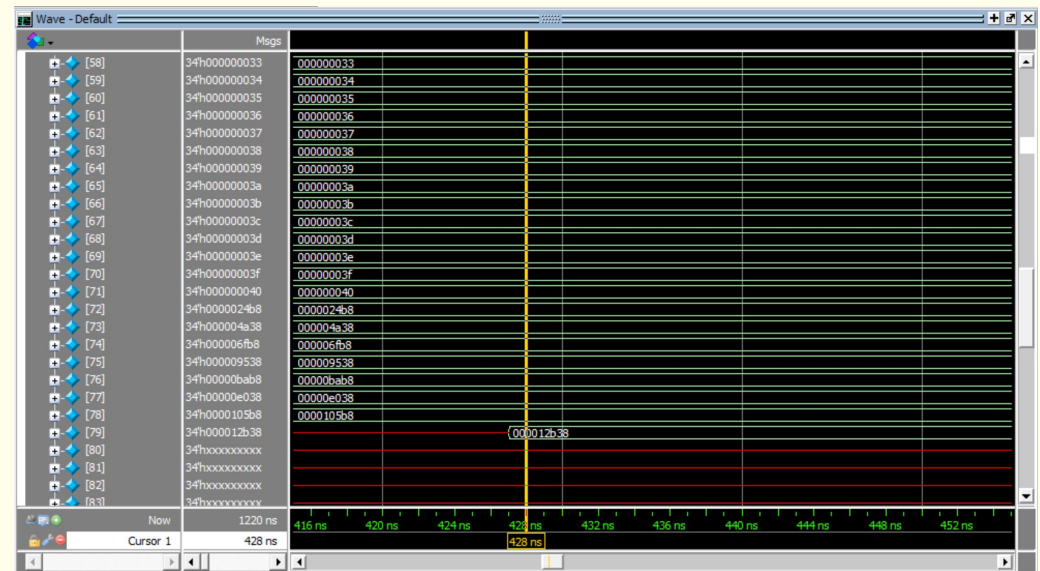
$$\begin{pmatrix} 1 \\ -4 \\ 6 \\ 3 \end{pmatrix} \begin{pmatrix} 2 \\ 6 \\ -1 \\ 7 \end{pmatrix} = -7$$

Results

Mem (2)



Mem (1)



Mem (3)

